# Week 12: International Climate Finance

FINA3020 at CUHK Business School

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# Lecture Overview

[Core Concept]

- International climate finance architecture, UNFCCC commitments, and multilateral institutions
- Concessional climate finance instruments: grants, loans, and guarantees
- Blended finance mechanics: risk allocation, additionality, and private-capital mobilization
- Just Energy Transition Partnerships (JETPs) and coal phaseout platforms
- Green bonds, greenium pricing, and empirical greenwashing evidence
- Bank net-zero commitments and credit allocation (Sastry, Verner, & Marques-Ibanez 2026)
- Compliance carbon markets: EU ETS, China CNCM, and Brazil CBIO
- Voluntary carbon offset markets and verification challenges
- In-class game: Blended-Finance Allocation

# International Climate Architecture

[Core Concept]

Climate finance supports mitigation and adaptation across countries.

- **UNFCCC finance goals**: The prior developed-country goal was US$100 billion annually through 2025. The COP29 NCQG calls for at least US$300 billion annually by 2035, led by developed countries, within a broader US$1.3 trillion mobilization effort from all sources.
- **Green Climate Fund (GCF)**: An operating entity of the UNFCCC financial mechanism using grants, concessional loans, equity, and guarantees.
- **Multilateral Development Banks (MDBs)**: Institutions such as the World Bank, ADB, and AfDB lend, invest, guarantee, and help prepare projects. Products differ across each group.

**Official Development Assistance (ODA)**

- Bilateral agencies and development banks, including JICA and KfW, provide grants and loans.
- Focus on adaptation, capacity building, and policy reform.

**Private Capital Linkage**

- Export credit agencies insure or guarantee trade, often with national-export objectives.
- Blended structures connect institutional ESG funds to emerging market projects.

Institutions and historical commitments →

# Concessional Finance and Externality Targeting [Core Concept]

Concessional climate finance provides a grant element or risk-bearing terms more favorable than comparable commercial finance. Providers can be public or philanthropic.

- Involves reduced interest rates, extended maturities, grace periods, or first-loss risk coverage.
- **Targeting Market Failures**: Can support projects where social returns exceed captured private returns, or where credit, information, coordination, and public-good problems block investment.

**Economic Externalities**

- Technology learning-by-doing and cost reduction spillovers.
- Early-stage project preparation and local grid infrastructure network effects.

**Environmental Externalities**

- Underpriced carbon emissions and unpriced global warming damages.
- Adaptation can create public and distributional benefits, but underinvestment can also reflect fiscal constraints, incomplete insurance, and difficulty monetizing avoided losses.

# Concessional Instrument Categories [Core Concept]

Public climate finance uses several instruments whose reported shares depend on the dataset and whether flows are counted at face value or grant equivalent:

- **Concessional Loans**: Used for revenue-generating or fiscally supported investments. Their subsidy is the grant element, not the full face value.
- **Direct Grants and Subsidies**: Useful for technical assistance, project preparation, adaptation, public goods, and settings with little repayment capacity.
- **Risk Guarantees and Tranching**: Credit enhancement structures absorbing initial project defaults to mobilize private commercial capital.

Grants are suited to activities with weak cash flows. Loans spread costs over time but add debt. Guarantees transfer specified risks and create contingent public liabilities. Instrument choice should target the actual market failure and avoid unnecessary subsidy.

# Concessional Loans, Guarantees, and Grants [Core Concept]

- A concessional loan offers financing below an appropriate market benchmark. Its subsidy depends on interest, maturity, grace periods, and credit risk, rather than its full principal.
- A guarantee covers specified losses up to a contractual limit. It transfers risk to the guarantor and can create a contingent public liability.
- A grant provides funds without loan repayment. It can support project preparation, adaptation, technical assistance, and activities with weak private cash flows.
- Public institutions, development banks, and philanthropic or private providers can supply concessional capital. The optimal instrument depends on the underlying market failure.

# Global Concessional Climate Finance Data

Empirical patterns documented by Climate Policy Initiative (2024):

- Concessional debt dominates mitigation financing, while grants concentrate in adaptation.
- The chart describes domestic concessional finance, not all domestic climate finance. Lower-income economies also depend on international support.

CPI statistics and definitions →

# Global Concessional Climate Finance Data

[Background Context]

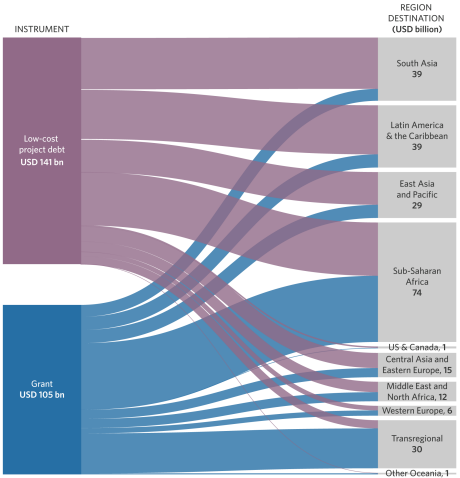
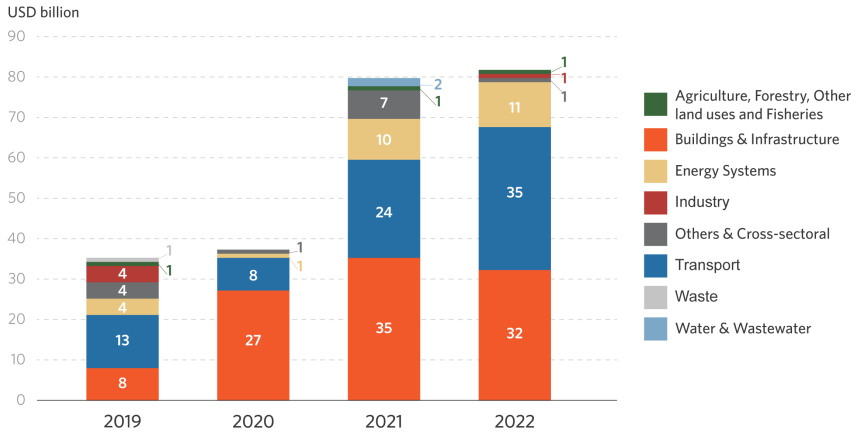


**Figure 3.8**: Domestic concessional climate finance by sector (2019-2022; USD billion)



# Blended Finance Mechanics

[Core Concept]

**Blended finance** combines public or philanthropic concessional funds with private commercial capital.

- **Core Objective**: Adjusts the allocation of risk or return so a project can attract commercial capital while preserving development additionality.
- **First-Loss Tranches**: Public or philanthropic capital absorbs losses before senior private claims, transferring rather than eliminating project risk.
- **Mobilization Ratio**: Private finance attributed to the intervention per dollar of public finance or subsidy. Definitions vary, and a high ratio alone does not prove additionality or social value.

**Sucden Cocoa Example (Ivory Coast)**

- USD 100M cocoa working-capital facility.
- USD 40M concessional support from GAFSP and IDA-PSW, administered by IFC.
- Remaining financing and mobilization should not all be classified as private without checking providers.

Sucden: verified financing terms →

**Market Scale Constraint**

- Reported blended-finance volumes are small relative to estimated climate and development investment needs.
- Comparisons depend on whether one counts commitments, disbursements, face value, or grant-equivalent subsidy.

# Grant Equivalent and Private Capital Mobilization [Core Concept]

Hypothetical one-year loan of USD 100 at 2%, compared with an appropriate 8% market benchmark. Ignore default and fees.

$$\text{Grant equivalent} = 100 - \frac{102}{1.08} = 5.56.$$

- The loan's face value is 100, while its modeled concession is 5.56. Finance totals measured at face value and grant equivalent are not interchangeable.
- If USD 20 of concessional capital accompanies USD 80 of commercial capital, commercial-to-concessional leverage is 4 times, while total-to-concessional leverage is 5 times.
- Mobilization is not automatically additionality. Ask whether the commercial investment would have occurred without the support.

Grant-equivalent example: all steps →

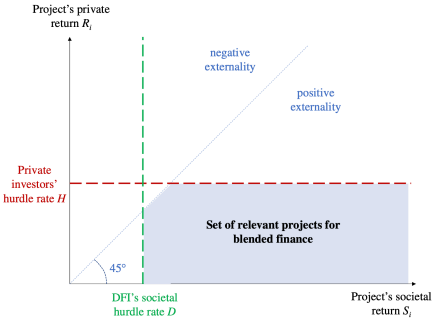
# Hurdle Rates and Risk Tranching

**[Core Concept]**

Commercial investors generally require risk-adjusted returns, while development institutions may accept concessional returns in exchange for verified social additionality.

- Green projects generate social benefits $S$ that private revenues do not capture.
- Risk, weak cash flows, transaction costs, or policy uncertainty can leave a project below the commercial hurdle rate.
- First-loss capital subordinates public funds and can lower expected senior loss. Total project risk remains and must be priced or borne by the junior tranche.

Tranche Payoffs → Capital Split →

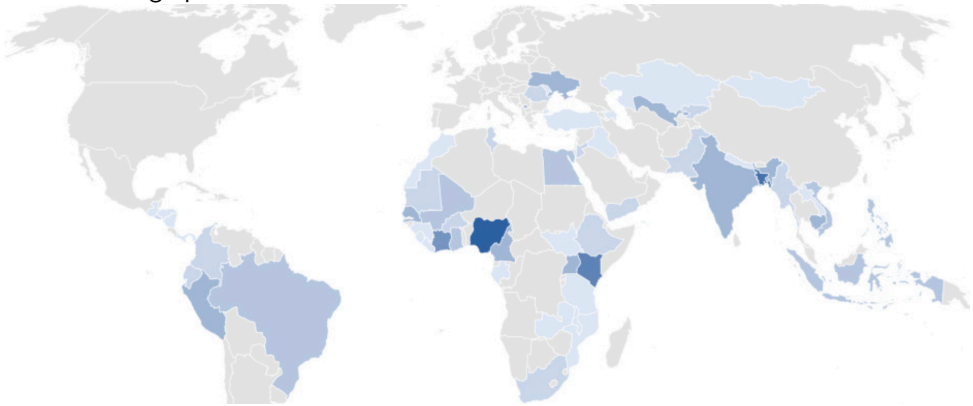



Source: Flammer, Giroux, and Heal (2025)

# Global Map of Blended Finance Deals



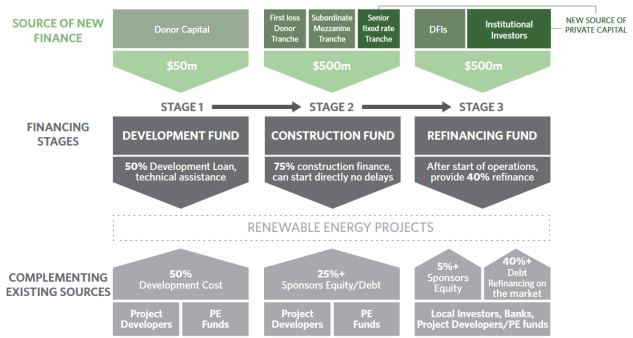
Geographic distribution of cross-border blended finance transactions:



Source: Flammer, Giroux, and Heal (2025) using IFC Data

- Darker shading indicates higher deal volume, concentrated across Sub-Saharan Africa, South Asia, and Latin America.
- The pattern is consistent with a role for country and market risk, but a deal map alone cannot show that private investment was impossible or that public finance caused mobilization.

Climate Investor One (CI1) manages renewable energy capital across development and construction stages.




Source: Climate Policy Initiative

- **Development Fund**: USD 56M donor grant capital funding feasibility studies.
- **Construction Equity Fund**: USD 863M multi-tiered capital structure.
- **Tranche Architecture**: Donor and concessional junior capital take more risk than commercial investors across development and construction facilities. Exact claims differ by fund.

# Different Structures for Blended Finance

[Core Concept]

- Climate Investor One combines financing across development and construction stages, including subordinated capital and guarantees. The risk allocated to each provider depends on the tranche and stage.
- The Responsible Commodities Facility supports eligible sustainable agriculture through a different financing structure. A public institution need not be the sole source of concession.
- A loan involving a development bank is not automatically concessional. IFC explicitly discloses concessional support and an estimated subsidy for the Sucden facility. Face value and subsidy value differ.
- Compare expected public loss, project additionality, and environmental outcomes. A high private-finance ratio alone is not evidence of a better project.

Original blended-finance case details →

# Just Energy Transition Partnerships (JETPs)

[Background Context]

JETPs are government financing partnerships for a country's power-sector transition, including worker support. Coal retirement is one possible component.

- Announced packages mix grants, concessional and market-rate loans, guarantees, and private finance for grids, clean generation, policy reform, and worker/community support.
- Address stranded asset risk and energy transition labor displacement.

**Announced Initial Packages**

- **South Africa (2021)**: US$8.5 billion pledged by EU, UK, US, France, and Germany.
- **Indonesia (2022)**: US$20 billion public-private financing package.
- **Vietnam (announced 2022, investment plan 2023)**: US$15.5 billion package for power-sector transition.

**Implementation Challenges**

- Pledge totals are not disbursements. Terms, additionality, and fiscal implications differ across components.
- Complex negotiations over early coal PPA (power purchase agreement) termination.

# JETPs: Pledges, Investment Plans, and Delivery [Core Concept]

- South Africa's partnership was announced in 2021 with an initial USD 8.5B pledge. Indonesia's 2022 partnership announced USD 20B.
- Vietnam's partnership was announced in December 2022 with an initial USD 15.5B commitment. These are dated initial pledges, not totals disbursed or current membership lists.
- Financing combines instruments with different borrowing costs and conditions. Grid investment, coal retirement, worker support, and implementation capacity affect results.
- A carbon price addresses emissions incentives. Transition finance can address capital and distributional constraints. Their effects can complement each other.

# Green Bonds and Greenium Pricing

[Core Concept]

A **green bond** is a fixed-income instrument whose net proceeds are designated for eligible green projects under a stated framework.

- The ICMA Green Bond Principles are voluntary process guidelines covering use of proceeds, project evaluation, management of proceeds, and reporting. Regulation and taxonomies may add legal requirements.

- **Greenium**: A positive greenium is a lower yield relative to an otherwise comparable conventional bond. It is an estimated pricing difference, not guaranteed for every issue.

### Empirical Findings

- Caramichael and Rapp (2024): Issuance yield discount of 3 to 8 bps for green corporate bonds (2014–2021 sample).

- Greenium was prominent in late 2010s but has narrowed as green bond supply expanded.

### Catastrophe Bonds Distinction

- Cat bonds transfer specified disaster risk to investors. Principal or coupons are reduced only if contractual triggers are met.

- Green bonds fund standard capital investment with standard corporate/sovereign credit risk.

Greenium Mechanics → Financing Savings →

Prominent green bond issuances illustrate sovereign digital infrastructure and corporate use-of-proceeds financing:

- **Hong Kong SAR Digital Green Bonds**: The government has issued tokenized green bonds in multiple currencies and demonstrated digitally native issuance, settlement, and servicing.
  - Tokenization is a record-keeping and workflow design. It does not by itself mean settlement occurs in CBDC or remove legal, custody, and interoperability risk.
  - External reviews and a green framework support reporting, but dealer or structuring-bank participation is not independent verification of environmental outcomes.
- **Apple Green Bonds**: Over US$4.7 billion issued since 2016 to fund low-carbon aluminum smelting, solar data centers, and closed-loop recycling.

2025 digital green-bond terms →

# Green Bonds, Sustainability-Linked Debt, and Catastrophe Bonds [Core Concept]

- A green use-of-proceeds bond allocates funds to eligible environmental activities. Investors still face the issuer's ordinary credit and market risks unless the contract says otherwise.
- A sustainability-linked bond changes terms when specified performance targets are met or missed. It need not ring-fence proceeds in the same way.
- A catastrophe bond transfers specified disaster risk to investors through contingent loss of principal or interest. A green label alone does not do this.
- Apple's green-bond program illustrates corporate use of proceeds. Hong Kong's November 2025 digital green-bond issue illustrates that currency, environmental use, and tokenized settlement are separate dimensions.

# Greenwashing and Moral Hazard

[Core Concept]

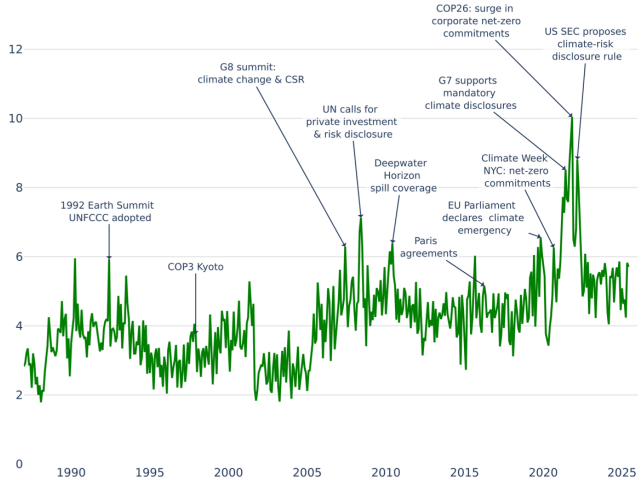
**Greenwashing** occurs when an entity misrepresents environmental performance or misallocates green bond proceeds.

- Post-issuance emissions or investment can diverge from labels, but correlation with issuance does not by itself prove that a greenium caused moral hazard.
- Media- or NLP-based greenwashing indices measure allegations or gaps between claims and observed actions. They are noisy proxies, not direct proof of misallocation.

Source: Gourier and Mathurin (2025)

# Greenwashing and Moral Hazard

**[Core Concept]**

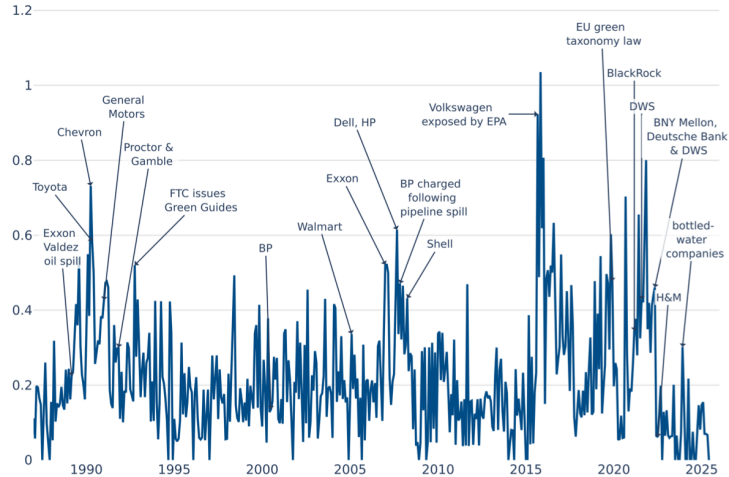



Firm Sustainability Index

Source: Gourier and Mathurin (2025)

# Greenwashing and Moral Hazard

**[Core Concept]**




Greenwashing Accusation Index

Source: Gourier and Mathurin (2025)

# Greenwashing and the Interpretation of Evidence



- Use-of-proceeds verification asks where funds went. Outcome evaluation asks whether emissions, adaptation, or other environmental measures improved.
- Leung, Wan, and Wong examine environmental performance around green issuance. Such evidence concerns their sample and method, rather than proving every issuer greenwashes.
- Gourier and Mathurin construct news-based sustainability and greenwashing measures. A news accusation index measures reported allegations, not independently verified emissions.
- Issuance selection, changing disclosure, reporting lags, and comparison groups matter. A temporary improvement before borrowing can create incentives for misleading signals.

Greenwashing: empirical findings →

# Bank Net-Zero Commitments and Portfolio Selection [Core Concept]

Many large banks have announced net-zero targets through voluntary alliances or their own transition plans. Membership and target coverage change over time.

- **Research Question** (Sastry, Verner, and Marques-Ibanez 2026): Do voluntary bank net-zero targets alter real credit supply, or reflect passive shifts in credit demand?
- **Study Setting**: Sastry, Verner, and Marques-Ibanez (2026) use European credit-registry data matched to emissions and transition-plan information. Results describe that sample and period.

## Ex-Ante Selection Patterns

- Committing banks are larger and differ from non-committers before treatment, so simple cross-sectional comparisons are not causal.
- Their ex-ante portfolios can be more carbon intensive, making selection into commitments and target choice central to identification.

## Reputational ESG Gains

- The study reports changes in ESG ratings around pledges, but ratings are assessments rather than emissions outcomes.
- A rating response can reflect information or investor demand. It does not establish that banks intentionally captured a benefit without changing lending.

# Testing Credit Supply vs. Demand in Net-Zero Banking [Core Concept]

Triple-differences design with firm-time fixed effects $\delta_{f,t}$ to absorb shocks common to a borrower at a given date (Khwaja and Mian 2008):

$$\ln(\text{Lending}_{b,f,t}) = \alpha_{b,f} + \delta_{f,t} + \gamma_{b,t} + \beta_1 \cdot \left(\text{PostNZ}_{b,t} \times \text{SectoralTarget}_{b,f}\right) + \varepsilon_{b,f,t}$$

- **Channel 1: Divestment**: The study finds no economically large relative decline in targeted-sector lending after commitment. Quoted rates fall modestly. Price and quantity signs alone do not identify a demand shift.
- **Channel 2: Green Project Finance**: Lending to some renewable projects rises, but pre-trends and tighter contract terms limit attribution of the increase to the pledge itself.
- **Channel 3: Borrower Engagement**: The study finds little detectable change in borrower target adoption or verified direct emissions over its horizon.

**Aspirations vs. Binding Commitments:** In this European sample and horizon, voluntary commitments do not appear to impose large binding portfolio constraints. Wider claims about transition plans or borrower demand require additional evidence.

Price-Quantity Credit Market Identification →

# Compliance Carbon Markets (CCMs)

[Core Concept]

Compliance carbon markets impose mandatory emissions obligations and allow regulated entities to trade eligible allowances or credits. A cap-and-trade ETS is one important design.

- **Cap-and-Trade ETS**: Regulator caps covered emissions $Q_{cap}$ and auctions or freely allocates tradable allowances.
- Regulated firms surrender one allowance per covered tonne of $CO_2e$, subject to the system's scope and compliance period.
- Firms with low marginal abatement costs abate more and sell allowances to firms with high marginal costs, establishing an allowance price $P_c$.

With accurate monitoring, enforceable compliance, competitive trading, and low transaction costs, a binding cap equalizes marginal abatement costs and achieves the covered target at least cost. Banking, market power, overlapping policies, and uncertainty can alter this benchmark.

Tax vs. Cap → | Two-Firm Example → | Allowance Trades →

Lecture Overview | Climate Architecture | Blended Finance and Risk Allocation | Green Private Capital | Carbon Markets | Participation Game | Backup Slides

# Compliance Carbon Market Design Attributes [Core Concept]

Key structural parameters governing compliance carbon credit markets:

- **Price and Quantity Controls**: Auction floors, cost-containment reserves, allowance banking, or supply-adjustment reserves can limit extreme outcomes. Their rules differ by system.
- **Sectoral Scope**: Begins with power generation and heavy industry before expanding to transport and buildings.
- **Holding Limits**: Limits position concentration, which reduces the risk of market manipulation.
- **External Offsets**: Limits maximum share of compliance obligations met through verified external offset credits (California's historical 4% limit applied in 2021–2025, limits depend on the compliance period).

# Allowance Markets, Offsets, and Policy Design [Core Concept]

- An allowance authorizes emissions under a compliance scheme. An offset credits a reduction or removal relative to a project baseline. They are not interchangeable without scheme rules.
- An absolute cap constrains total covered emissions. An intensity benchmark links the allocation to output, so emissions can grow when output grows.
- Auctions, free allocation, banking, reserves, price floors, holding limits, and offset limits distribute costs and affect investment incentives.
- Retiring an allowance or offset has effects that depend on supply adjustments, additionality, leakage, and permanence. CORSIA is an aviation offset framework, not simply another economy-wide cap-and-trade market.

# EU Emissions Trading System (EU ETS)

[Background Context]

Launched in 2005, the EU ETS covers roughly 40% of EU emissions across power, industry, aviation, and, since 2024, maritime transport within its defined scope.

- **Cap Dynamics**: Total emissions cap declines annually via the Linear Reduction Factor.
- **Market Stability Reserve (MSR)**: Adjusts auction supply using rules tied to the total number of allowances in circulation. It is not a discretionary price peg.
- **Distributional Impacts**: Carbon costs can pass through to energy prices, while the burden and innovation response depend on free allocation, revenue recycling, market structure, and household exposure.

Market scope and original empirical evidence →

Launched in 2021 for power generation, China's National Carbon Market expanded in 2025 to steel, cement, and aluminum smelting and is the world's largest by covered emissions.

- **Transitioning Design**: Output-based benchmark allocation makes the effective obligation intensity-based. Policy plans call for a gradual shift toward stronger absolute-quantity control in suitable sectors.
- **Flexibility Focus**: Accommodates rapid power demand expansion while incentivizing thermal efficiency improvements.
- **Evaluation Challenge**: Expansion, free allocation, changing benchmarks, and concurrent industrial policies make causal effects on innovation and productivity difficult to isolate.

# Brazil's CBIO (RenovaBio Program)



RenovaBio is Brazil's national biofuel policy, using tradable CBIO carbon credits to incentivize decarbonization.

- **Supply**: Certified biofuel producers generate CBIO credits based on lifecycle emission reductions evaluated by RenovaCalc.
- **Demand**: National targets are set by Brazil's CNPE and translated by ANP into mandatory distributor targets. Distributors acquire and retire CBIOs, with sanctions for non-compliance.
- **Market Structure**: CBIOs are book-entry environmental assets traded through B3 market infrastructure. Obligated distributors retire credits for compliance.

RenovaCalc → CBIO Revenue →

Voluntary carbon markets let buyers purchase credits from projects such as forestry, methane abatement, clean cooking, or carbon removal outside a mandatory buyer compliance obligation.

- **Quality Reassessment**: Confidence and traded value weakened after rapid growth as buyers scrutinized additionality and over-crediting. Issuance and retirement can follow different paths.
- **Additionality**: Difficulty proving the project would not have occurred without offset revenue.
- **Permanence and Leakage**: Wildfires risk reversing forest carbon storage. Protecting one forest tract may shift logging to adjacent land.

Offset quantities and verification example →

**Question: can a USD 20M public budget make a USD 80M loan acceptable?**

- Project cost: USD 100M = 20M sponsor equity + 80M private loan.
- A separate public budget funds a first-loss guarantee reserve, interest subsidy, or preparation grant. It does not replace the 80M loan.
- Baseline: default probability 10%; loan loss in default before the guarantee 60M; expected loan-loss limit 2%.
- Stress: probability 15%; loan loss 70M; expected loan-loss limit 2.5%.

All amounts are USD millions over one illustrative project period. Losses are positive. Ignore interest and discounting. Given loan losses already account for sponsor equity.

# In-Class Game: Blended-Finance Allocation

[Core Concept]

1. **Understand**: trace the example allocation: 12M guarantee, 5M subsidy, 3M grant.
2. **Practise**: budget, loss waterfall, expected loss, loss rate, required guarantee, and mobilization. Hints and worked solutions are always available.
3. **Decide**: allocate up to 20M independently in baseline and stress.
4. **Compare**: revisit choices; inspect feasible alternatives, waterfall, and exact two-state distribution. No repeated arithmetic or random draws.
5. **Reflect**: explain feasibility and a proposed change; then enter identity and submit. Wait for a server receipt.

Student activity | Standalone student explanation

# Worked Example: Trace the Loan Loss

[Core Concept]

Use the baseline and a 12M guarantee, 5M subsidy, and 3M grant.

1. Budget used: $12 + 5 + 3 = 20$M.
2. In default: public payout $= \min(60, 12) = 12$M; private loss $= \max(60 - 12, 0) = 48$M. Total loss remains 60M.
3. Expected private loss: $0.90(0) + 0.10(48) = 4.80$M. Expected public payout: $0.10(12) = 1.20$M.
4. Expected private loss rate: $100(4.80/80) = 6.00\%$, above the 2% limit. The lender declines.

The 12M reserve is committed capacity; 1.20M is its expected payout. Neither is an investment return. Round displayed dollars and rates to two decimals.

Open worked example

# How Much Guarantee Would Be Enough? [Core Concept]

Let $G$ be external guarantee capacity; loan principal remains 80M.

$$\underbrace{p\max(D-G,0)}_{\text{expected private loss}} \le x(80) \quad \Rightarrow \quad G_{\min} = \max\{0, D - 80x/p\}.$$

- Baseline: allowed expected loss $= 0.02(80) = 1.60$M. Allowed default loss $= 1.60/0.10 = 16$M. Required guarantee $= 60 - 16 = 44$M.
- Stress: $G_{\min} = 70 - 0.025(80)/0.15 = 56.6667$M. At two decimals, 56.67M is sufficient.
- Both exceed the 20M budget. The best feasible guarantee cannot close the loan.

Subsidy and grant effects on risk are held fixed in this exercise. Investigate those channels before treating this as a real financing recommendation.

# Game Debrief: Compare Feasible Alternatives

[Core Concept]

| Expected private loan-loss rate | Baseline | Stress |
|---|---|---|
| Example: 12M guarantee / 5M subsidy / 3M grant | 6.00% | 10.88% |
| All 20M in guarantee | 5.00% | 9.38% |
| All subsidy, all grant, or no support | 7.50% | 13.13% |
| Lender limit | 2.00% | 2.50% |

- Waterfall: public payout plus private loss equals loan loss in the selected state. With no default, both are zero.
- Distribution: baseline 90% no default / 10% default; stress 85% / 15%. The expected value averages those losses.
- No feasible package attracts the lender: actual mobilization is 0. A no-default outcome would not establish that a policy was better.

Comparison | Waterfall | Distribution

# Game Debrief: What Would Make the Project Feasible? [Core Concept]

- **Recommendation**: explain your loss rates relative to the limits. Use the all-guarantee comparison and the required protection.
- **Possible changes**: investigate lower default risk or loss severity, a different financing structure, or additional justified support. Declining the project is an available recommendation.
- **Mobilization**: 80M private / 20M public = 4 times only if the loan closes. In this game it does not. Total project/public support = 5 times is a different ratio.
- **Participation**: review six calculations (attempt or worked solution), use feasible budgets, view comparisons, and write a reflection. Correct guesses and project success are not requirements. A local draft is not a server receipt.

Extension: internal junior capital → Extension: changing senior principal →

# Appendix: Formal Risk-Tranching Payoff Model

[Core Concept]

For a one-period principal-only illustration, let initial project capital be $K = F + S$, where $F$ is junior first-loss capital and $S$ is senior debt. At maturity, project value is $\tilde{V} = K - L$ for loss $0 \leq L \leq K$. Payoffs at maturity:

1. Senior principal payoff: $y_S = \min\{S, \max(0, \tilde{V})\}$
2. Junior payoff: $y_F = \max\{0, \tilde{V} - S\} = \max\{0, F - L\}$

Junior loss is $L_F = \min\{L, F\}$. Senior loss is $L_S = \min\{S, \max(0, L - F)\}$. Tranching reallocates $L$. It does not reduce total project loss unless the public intervention also changes project cash flows or default risk.

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# Appendix: Cap-and-Trade vs Carbon Tax Mechanics [Core Concept]

Let $MC_i(a_i)$ denote marginal abatement cost for firm $i$ reducing emissions by $a_i$:

1. **Carbon Tax** $t$: Firm chooses abatement where $MC_i(a_i) = t$. Gives price certainty but quantity uncertainty.

2. **Cap-and-Trade**: Regulator sets aggregate cap $\bar{Q} = \sum(E_i^0 - a_i)$. Permit market clearing yields equilibrium allowance price $p^* = MC_1(a_1^*) = MC_2(a_2^*)$.

In the canonical Weitzman (1974) approximation with uncertainty and no hybrid adjustment, a quantity instrument is relatively attractive when marginal damages are steep relative to marginal abatement costs. A price instrument is relatively attractive when damages are flatter. Real systems add banking, collars, and other hybrids.

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# Appendix: Greenium Yield Spread Mechanics

[Core Concept]

Let $y_{conv}$ and $y_{green}$ denote yields on otherwise comparable conventional and green bonds (ideally same issuer, currency, seniority, maturity, and liquidity):

$$y_{green} = y_{conv} - \delta_{green}$$

where $\delta_{green} = y_{conv} - y_{green}$ is positive for a greenium but can be zero or negative. A positive spread can reflect investor preferences, mandates, scarcity, liquidity, tax, certification, or index effects. Greater green-bond supply can narrow a scarcity premium, but $\delta_{green}$ need not converge mechanically to zero because bond characteristics and investor segmentation also change.

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# Appendix: Price-Quantity Identification Matrix in Credit Markets [Core Concept]

If only one curve shifts and all else is fixed, textbook price–quantity signs are:

| Hypothesized Strategy | Loan Quantity ($Q$) | Interest Rate ($r$) |
|---|---|---|
| **Credit Supply Reduction** | $Q \downarrow$ | $r \uparrow$ |
| **Credit Demand Reduction** | $Q \downarrow$ | $r \downarrow$ |
| **Credit Supply Expansion** | $Q \uparrow$ | $r \downarrow$ |
| **Credit Demand Expansion** | $Q \uparrow$ | $r \uparrow$ |

Observed signs do not identify a curve when supply and demand move together or loan composition changes. Firm-time fixed effects compare committed and uncommitted banks lending to the same multi-bank borrower at the same date and absorb common borrower demand. Interpreting the residual as bank supply still requires no bank-specific borrower-demand shock correlated with commitment.

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# Appendix: Brazil CBIO RenovaCalc Microstructure

[Core Concept]

In simplified form, eligible biofuel sales by producer $j$ generate CBIO issuance according to:

$$\text{CBIO}_j = \text{Eligible Volume}_j \times \text{NEEA}_j \times \eta_j$$

where $\text{NEEA}_j$ is the certified energy-environmental efficiency score derived from lifecycle assessment and $\eta_j$ converts physical units. One issued CBIO represents one tonne of $CO_2e$ avoided under the program methodology. Fuel distributors receive annual targets and retire CBIOs for compliance.

There is no valid universal penalty formula of "R$100,000 per missing CBIO plus twice the price": sanctions follow the governing statute and regulations, and an unmet quantity can also be carried into the next target period.

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# Climate Finance Institutions: Different Mandates and Instruments [Core Concept]

- The UNFCCC process establishes collective goals and reporting arrangements. The USD 100B goal originated in Copenhagen in 2009 and was later extended through 2025. It did not originate with the 2015 Paris Agreement.
- The Green Climate Fund and Global Environment Facility serve the UNFCCC financial mechanism. The Adaptation Fund supports adaptation projects and programs. Its institutional and funding arrangements differ.
- The Climate Investment Funds work through multilateral development banks. MDB loans, private-sector investments, guarantees, and grants involve different balance sheets and risk-bearing arrangements.
- Mitigation lowers emissions. Adaptation reduces damage from climate exposure. Grants may be especially useful when benefits are public or hard to monetize. Revenue-generating projects may support repayable finance.

Commitments, approvals, disbursements, and measured environmental outcomes describe different stages. A pledge is not cash already spent.

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**CPI's 2024 report, international concessional flows in 2019–2022:**

| Instrument | Share of finance | Average project size |
|---|---|---|
| Concessional loans | 57% | USD 74M |
| Grants | 43% | USD 3M |

- Finance shares weight dollars. Average project sizes divide funding by project counts. Many small grants can coexist with a larger loan share by value.
- A face-value loan total includes principal expected to be repaid. A grant-equivalent measure instead estimates the value of the concession.
- Domestic concessional flows in the report are a separate category from international flows and from the broader climate-finance market.

Source: CPI, Understanding Global Concessional Climate Finance 2024.

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# Sucden Cocoa Financing: The Disclosed Transaction Terms [Core Concept]

IFC's disclosure describes up to USD 100M for cocoa procurement and export in Côte d'Ivoire: a one-year working-capital loan, renewable for up to four years.

1. Concessional support totals USD 40M: USD 10M from GAFSP and USD 30M from the IDA Private Sector Window, administered by IFC.

2. The disclosure estimates subsidy at 1.2% of the USD 100M project cost: $100\text{M} \times 0.012 = \$1.2\text{M}$. The concession is not the full 40M principal.

3. The overall financing includes IFC and mobilization/parallel financing. Subtracting 40M from 100M does not by itself establish 60M of private finance.

4. Working capital pays for cocoa before export receipts arrive. This links climate/development financing back to Week 7's trade-finance timing problem.

Source: IFC, Sucden CDI project 47216.

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# Grant Equivalent and Mobilization: Compute Two Different Measures [Core Concept]

## Grant equivalent of a cheap one-year loan

1. Advance USD 100 today at 2%. Borrower repays 102 in one year.
2. Discount at the appropriate assumed market benchmark of 8%:

$$PV(\text{repayment}) = 102/1.08 = 94.4444.$$

3. Concession $= 100 - 94.4444 = 5.5556$, or 5.56% of face value. The 6 percentage point rate gap is not itself a 6% present-value grant.

## Capital mobilization ratios

Public/concessional capital 20 and private capital 80 finance a 100 project.
Private/public $= 80/20 = 4$. Total/public $= 100/20 = 5$.
Neither ratio establishes how much private finance is *additional* to what would otherwise have occurred.

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# Two Blended Structures: Who Provides the Concession?

[Core Concept]

## Responsible Commodities Facility

UK retailers including Tesco, Sainsbury's, and Waitrose provided subordinated concessional debt supporting deforestation-free soy supply chains. Foundation and corporate grants helped design and test the structure. Private actors can accept below-market returns for strategic supply-chain benefits.

## Climate Investor One

CPI reports USD 56.2M of donor capital in the development fund and USD 863.5M in construction equity. First-loss and subordinate capital, plus an export-credit guarantee, support more senior investors. Its reported fourfold commercial mobilization per donor dollar is a dated program measure, not a universal multiplier.

Development support can create investable projects. Construction capital absorbs a different set of risks. Follow each tranche's contractual claims.

Source: CPI 2024, case studies, pp. 16–19.

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# Hong Kong Digital Green Bonds: November 2025 Issuance

[Background Context]

| Currency | Principal | Tenor | Annual coupon |
|---|---|---|---|
| HKD | 2.5 billion | 2 years | 2.500% |
| RMB | 2.5 billion | 5 years | 1.900% |
| USD | 300 million | 3 years | 3.633% |
| EUR | 300 million | 4 years | 2.512% |

- Announced on 11 November 2025, the four tranches totaled approximately HKD 10B equivalent. The HSBC Orion platform supported issuance and CMU settlement infrastructure.
- HKD and RMB tranches offered primary settlement using tokenized central-bank money, including e-HKD and e-CNY options.
- Tokenized issuance and tokenized payment are distinct features. This specific deal included both options. Neither feature guarantees the environmental outcome.

Source: HKSAR Government, 11 November 2025 announcement.

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# Greenium: Translate Basis Points into Financing Savings

[Core Concept]

Illustration: comparable five-year conventional and green bonds are issued at par, with annual yields/coupons of 5.00% and 4.95%, on USD 100M principal.

1. Define greenium as conventional minus green yield: $5.00\% - 4.95\% = 0.05\% = 5$ bps.
2. Annual coupon saving is $100\text{M} \times 0.0005 = \$50{,}000$.
3. At a common 5% discount rate, five-year PV of savings is $50{,}000 \sum_{t=1}^{5} 1.05^{-t} = \$216{,}474$.
4. Deduct extra certification, reporting, and other issuance costs before claiming a net benefit. Comparable credit risk, maturity, liquidity, and currency are essential.

Caramichael and Rapp's 2022 working paper estimates an average issuance spread advantage near 8 bps in its sample. This empirical estimate does not imply a permanent premium for all green bonds.

Source: Caramichael and Rapp, IFDP 1346.

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# Greenwashing Evidence: Restore the Finding and Its Limits



Leung, Wan, and **Wong** (2023) examine emissions intensity and subsequent issuance in corporate green-bond markets.

- About one-third of sampled issuers show worse environmental performance after their initial green issuance under the paper's measures.
- Firms classified as greenwashing are less likely to issue green bonds again and face higher issuance costs when they do.
- Well-defined taxonomies and stronger environmental disclosure are associated with less measured greenwashing.

**Interpretation**

Firm-wide emissions intensity imperfectly measures the use of particular proceeds. The results do not prove that cheaper green-bond funding caused the deterioration.

Source: HKIMR WP 01/2023, The Shade of Green.

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# Cap-and-Trade, Step 1: Find Efficient Abatement

[Core Concept]

Two firms each initially emit 100 tonnes. The cap is 140, so total abatement must be 60. Marginal abatement costs are $MAC_A = 2a_A$ and $MAC_B = 4a_B$, in dollars per tonne.

1. Binding cap: $a_A + a_B = 200 - 140 = 60$.
2. At a competitive interior solution, equalize marginal costs: $2a_A = 4a_B$, so $a_A = 2a_B$.
3. Substitute into the cap: $3a_B = 60$, giving $a_B = 20$ and $a_A = 40$.
4. Allowance price is $p = 2(40) = 4(20) = \$80$ per tonne. Remaining emissions are 60 at A and 80 at B.

The low-cost firm abates more. A carbon tax of \$80 would induce the same quantities under these known cost curves. Uncertainty can make price and quantity instruments differ.

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# Cap-and-Trade, Step 2: Costs and Allowance Trades

[Core Concept]

Abatement cost is the area under each marginal-cost curve:

$$C_A(a) = a^2, \qquad C_B(a) = 2a^2.$$

1. Efficient allocation costs $40^2 + 2(20^2) = 1{,}600 + 800 = \$2{,}400$.
2. Requiring 30 tonnes of abatement at each firm costs $30^2 + 2(30^2) = \$2{,}700$. Trading saves \$300.
3. If each firm initially receives 70 allowances, A emits 60 and sells 10. B emits 80 and buys 10. Payment is $10 \times 80 = \$800$.
4. A's abatement cost net of sales is $1{,}600 - 800 = 800$. B's cost including purchases is $800 + 800 = 1{,}600$. Their total is still 2,400.

Allowance payments redistribute costs. They are not extra resource costs on top of abatement. Initial allocation affects who pays, under the benchmark assumptions.

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# Carbon-Market Scope and the Distribution of Costs



- The EU ETS covers specified sectors and gases: chiefly $CO_2$, with nitrous oxide and perfluorocarbons from specified activities. ETS2 is a separate system designed for fuel emissions in buildings, road transport, and additional sectors.
- RGGI covers power-sector $CO_2$ in participating US states. California has broader coverage, an auction floor, and containment mechanisms. Offset eligibility and limits are system- and period-specific.
- Känzig's paper, revised in September 2025, identifies EU carbon-policy shocks using institutional events and high-frequency data. Restrictive shocks raise energy prices, reduce emissions, and spur green innovation, but depress activity in the estimated response.
- Lower-income households face larger burdens through both energy spending and income/employment effects. Revenue recycling and complementary policy can change distributional outcomes.

Source: Känzig, The Unequal Economic Consequences of Carbon Pricing.

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# CBIO: Convert a Lifecycle Score into Credit Revenue

[Core Concept]

Illustrative eligible, certified biofuel volume: 1M liters. Energy content 20 MJ/liter. Certified avoided lifecycle emissions 50 grams $CO_2e$/MJ. Ignore eligibility adjustments for this exercise.

1. Energy supplied: $1\text{M liters} \times 20\ \text{MJ/liter} = 20\text{M MJ}$.
2. Avoided emissions: $20\text{M MJ} \times 50\ \text{g/MJ} = 1$ billion grams $= 1{,}000$ tonnes $CO_2e$.
3. At one tonne per CBIO, this yields 1,000 credits. At an assumed BRL 80 per credit, gross revenue is BRL 80,000, or BRL 0.08 per liter.
4. Certification, eligible sales, and issuance costs affect net revenue. RenovaCalc and program rules determine actual credits. These inputs are hypothetical.

Eligible pathways include ethanol and other certified biofuels. Expected credit prices and distributor targets affect investment and production incentives, including sugar-versus-ethanol choices.

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# Offsets: Distinguish Credits Issued, Traded, and Retired [Core Concept]

- A project baseline estimates emissions without the project. If claimed avoided emissions are 1,000 tonnes but a credible counterfactual supports only 600, issuing 1,000 credits overstates the benefit by 400 tonnes.
- Additionality asks whether the activity needed credit revenue. Leakage asks whether emissions moved elsewhere. Permanence asks whether stored carbon can later be released.
- Issuance creates credits. Trading transfers them, potentially many times. Retirement takes them out of circulation for a claim. A fall in trading value does not necessarily mean retirements fell by the same percentage.
- Investigations, legal scrutiny, corporate-claim rules, and revised standards can alter demand and prices. Independent verification and conservative baselines address different parts of the problem.

A low credit price can reflect cheap genuine abatement, weak demand, or poor quality. Price alone cannot identify which explanation holds.

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# Extension: Internal Junior-Capital Waterfall

[Core Concept]

Separate internal model: project cost 100M = 20M public junior + 80M private senior. Public capital absorbs project losses first. Ignore interest and discounting.

| | Baseline | Shock |
|---|---|---|
| Probability of default | 10% | 15% |
| Project loss if default occurs | 60M | 70M |
| Junior loss if default occurs | $\min(20, 60) = 20$M | $\min(20, 70) = 20$M |
| Senior loss if default occurs | $60 - 20 = 40$M | $70 - 20 = 50$M |
| Expected senior loss | $0.10(40) = 4$M | $0.15(50) = 7.5$M |
| Expected loss / senior capital | $4/80 = 5\%$ | $7.5/80 = 9.375\%$ |

Check total expected losses: baseline $2 + 4 = 6$M, shock $3 + 7.5 = 10.5$M. Tranching reallocates the same project loss. It does not eliminate it.

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# Extension: Junior Capital Replaces Senior Principal

[Core Concept]

Keep project cost at 100M. Let public junior capital be $J$ and replace senior capital dollar for dollar, so senior capital is $100 - J$. Target expected senior principal loss no higher than 2%.

**Baseline: $p = 0.10$, default loss $L = 60$**

$$\frac{0.10(60 - J)}{100 - J} \leq 0.02 \;\Rightarrow\; 6 - 0.10J \leq 2 - 0.02J \;\Rightarrow\; J \geq 50.$$

At $J = 50$, senior capital is 50 and expected senior loss is 1: exactly 2%.

**Shock: $p = 0.15$, default loss $L = 70$**

$$0.15(70 - J) \leq 0.02(100 - J) \;\Rightarrow\; 8.5 \leq 0.13J \;\Rightarrow\; J \geq 8.5/0.13 \approx 65.385.$$

Both exceed 20M. Stress threshold: 65.384615M, or at least 65.39M at two decimals.

An external guarantee that leaves senior debt at 80M has different algebra. Specify whether support replaces debt or adds protection outside the capital structure.

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# A Concessional Tranche Can Bridge a Private-Return Shortfall [Core Concept]

Stylized certainty example: a project costs USD 100M and earns net income of 6M in one year, with principal preserved. Commercial investors require 8%. A concessional investor accepts principal repayment with zero income.

1. With only commercial capital, the return is $6/100 = 6\%$, below the 8% hurdle.
2. Let concessional capital be $K$. Commercial capital is $100 - K$ and receives the entire 6M income.
3. Solve $6/(100 - K) = 0.08$: commercial capital must be $6/0.08 = 75$M, so concessional capital is 25M.
4. General formula: $K = C(1 - R/H) = 100(1 - 0.06/0.08) = 25$M.

This return subsidy differs from a default-loss guarantee. Social benefits must justify the concession, and the investment must be additional.

Source: Flammer, Giroux and Heal, AEA Papers and Proceedings, 2025.

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